COURSE TITLE : BIO – CHEMICAL ENGINEERING
COURSE CODE : 6074
COURSE CATEGORY : E
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 60
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Bio-Process Fundamentals	15
2	Enzymes	15
3	Fermentation	15
4	Recovery and Purification of Products	15
TOTAL		**60**

COURSE OUTCOMES :

SL.NO.	SUB	STUDENT WILL BE ABLE TO
1	1	Know the bio-process fundamental
2	2	Understand the functions of Enzymes
3	3	Understand various fermentation processes
4	4	Understand the importance of downstream process steps in bio processing

SPECIFIC OUTCOMES

MODULE - I

BIO-PROCESS FUNDAMENTALS

1.1.0 Know the bio-process fundamentals

1.1.1. Define Biochemical Engineering.
1.1.2. Define micro biology.
1.1.3. List the applications of bio- technology
1.1.4. Describe integrated bio-process
1.1.5 Classify the organism
1.1.6. Draw the figure of simplified classification of ancestors.
1.1.7. Define bio-molecules
1.1.8. Explain in detail, the structure of Prokaryotic and Eukaryotic cells with figure.
1.1.9. Identify the micro organisms -Bacteria, Viruses, Fungi, Algae, and Protozoa.
1.1.10. Describe the classification, function and properties of Lipids.
1.1.11. Explain the function and properties of Fatty acids.
1.1.12. Describe the function and properties of steroids.
1.1.13. Explain separation and purification of Proteins.
1.1.14. Describe the properties and function of Vitamins.

1.1.15. Explain the properties and function of Carbohydrates.

1.1.16. Describe the function and properties of DNA and RNA.

MODULE - II

ENZYMES

2.1.0 Understand the functions of Enzymes.

2.1.1 Define enzyme.
2.1.2 Explain the function of Enzymes as biological catalyst.
2.1.3 List the terms associated with enzymology
2.1.4 Describe the chemical theories behind the functioning of enzymes
2.1.5 Define enzyme specificity.
2.1.6 Explain different types of Enzyme specificity.
2.1.7 Define active site of the enzyme.
2.1.8 Explain Fischer Lock and Key hypothesis in Enzymes.
2.1.9 Explain immobilization of Enzymes.
2.1.10 Name the important Enzyme immobilization methods.
2.1.11 Describe different methods of immobilization of Enzymes.
2.1.12 Explain the properties of immobilized Enzymes.
2.1.13 Explain the types of reactors used for Enzyme immobilization with a neat fig
2.1.14 Classify various type of ideal flow reactors
2.1.15 Classify the bio-reactors based on feeding mechanism.
2.1.16 Explain the various industrial application of enzymes

MODULE - III

FERMENTATION

3.1.0 Understand various fermentation processes

3.1.1 Define fermentation.
3.1.2 State the requirements of fermentation process.
3.1.3 Draw a schematic representation of fermation process.
3.1.4 Explain the constructional details of a fermenter with a neat sketch
3.1.5 Explain the controlling and monitoring parameters of fermenter.
3.1.6 Describe anaerobic and aerobic fermentation process.
3.1.7 Distinguish between anaerobic and aerobic fermentation process.
3.1.8 Describe solid state fermentation.
3.1.9 State the advantages of solid state fermentation.
3.1.10 Explain submerged fermentation .
3.1.11 List the advantages of sub merged fermentation
3.1.12 Differentiate between solid state fermentation and submerged fermentation.
3.1.13 List the applications of solid state and submerged fermentation.
3.1.14 State various types bioreactors
3.1.15 Explain fluidized bed reactors and air lift reactors
3.1.16 Describe the manufacture of ethyl alcohol by fermentation of molasses
3.1.17 Explain the production of antibiotics through fermentation.
3.1.18 Describe the manufacture of penicillin.

MODULE – IV

RECOVERY & PURIFICATION OF PRODUCTS

4.1.0 Understand the importance of downstream process steps in bio processing

4.1.1 List the various unit operations come across downstream processing
4.1.2 Define membrane
4.1.3 List the different types of membrane separation process
4.1.4 State the advantages and disadvantages of membrane separation process
4.1.5 List the polymeric materials that can be used as membranes
4.1.6 State the important types of membranes
4.1.7 State the characteristics of good membrane
4.1.8 Explain the different types of membrane module
4.1.9 List different types of Chromatographic process
4.1.10 Describe in detail the affinity chromatographic process
4.1.11 Explain the various application of Chromatographic separation process
4.1.12 Explain the Dialysis process and mention its applications.
4.1.13 Describe Microfiltration and its application
4.1.14 Explain Nanofiltration and its application
4.1.15 Explain ultra filtration and mention its application
4.1.16 Describe reverse osmosis process and mention its applications

COURSE CONTENT

MODULE I

Bio chemical Engineering – Introduction to Microbiology-structure of cells-cell theory-prokaryotic and eukaryotic cell- classification of micro organisms-bacteria, viruses, fungi, algae, protozoa- lipids –fatty acids – steroids- purification of proteins-properties and functions vitamins, carbohydrates, DNA, RNA

MODULE II

Enzymes-nomenclature and classification- Biological catalyst- Enzymes specificity-Active site-Enzyme immobilization. Different types reactors-ideal flow-feeding mechanism- properties and applications of immobilized enzymes.

MODULE III

Fermentation-requirements-schematic representations-fermenter –constuction details-controlling parameters-aerobic and anaerobic process-solid state fermentation-submerged fermentation-bioreactors- -manufacture of ethyl alcohol –penicillin.

MODULE IV

Downstream processes-membrane-types of membranes-separation processes- dialysis-chromatography-micro filtration-nano filtration- ultra filtration- reverse osmosis.

REFERENCES

James.E.Bailey and David.F.Olis	-	Biochemical Engineering Fundamentals
D.G Rao	-	IntroductiontoBiochemical engineering
Michael L.Shuler	-	Bioprocess Engineering
Sivasankaran	-	Bio-separation